

Xuwan Yin

• xuwan Yin@gmail.com • Qingdao, Shandong, China
• [LinkedIn](#) • [Github](#) • [Google Scholar](#) • (434) 327-0325

Education

2016 – 2022	Ph.D., Computer Engineering, University of Virginia, United States Ph.D. thesis: Generative Modeling With Adversarial Training Supervisor: Prof. Gustavo Rohde Research focus: adversarial machine learning, energy-based generative models, multimodal models, optimal transport in signal processing
2011 – 2014	M.Eng., Computer Engineering, University of Science and Technology Beijing, China Master thesis: Robust Text Detection in Natural Scene Images Supervisor: Prof. Xu-Cheng Yin
2007 – 2011	B.Eng., Computer Science and Engineering, University of Science and Technology Beijing, China

Employment

2025 – Present	Independent Researcher, San Francisco / Qingdao Scalable energy-based generative models. Developed an adversarial training framework that unifies generation and classification in a single model, the first to scale energy-based models to ImageNet 256×256 (ICLR 2026).
2023 – 2025	Research Engineer, Center for AI Safety, San Francisco, United States Research on LLM safety evaluation, automated red teaming, adversarial robustness, and emergent value systems in AI. Contributed to HarmBench (ICML 2024), Utility Engineering (NeurIPS 2025), Representation Engineering , and Unforeseen Adversaries .
2014 – 2015	Computer Vision Engineer, Yuanli Education Technology, Beijing, China Text detection and recognition in natural images.

Research Interests

Energy-based generative models

- Scalable training of energy-based models via adversarial training
- Text-to-image generation with energy-based models
- Hybrid discriminative-generative models

Machine learning robustness and security

- Training adversarially robust image models and LLMs
- Safety evaluation/red-teaming/jailbreaking LLMs
- Detecting adversarial examples

Selected Publications

- Scalable Energy-Based Models via Adversarial Training: Unifying Discrimination and Generation
Xuwan Yin, Claire Zhang, Julie Steele, Nir Shavit, Tony T. Wang
International Conference on Learning Representations (**ICLR 2026**) [[arxiv](#)]

- Utility Engineering: Analyzing and Controlling Emergent Value Systems in AIs [\[code\]](#)[\[arxiv\]](#)
Mantas Mazeika, **Xu Wang Yin**, Rishub Tamirisa, Jaehyuk Lim, Bruce W. Lee, Richard Ren, Long Phan, Norman Mu, Adam Khoja, Oliver Zhang, Dan Hendrycks
NeurIPS 2025
- RenderAttack: Hundreds of Adversarial Attacks Through Differentiable Texture Generation
Dron Hazra, Alex Bie, Mantas Mazeika, **Xu Wang Yin**, Andy Zou, Dan Hendrycks, Maximilian Kaufmann
NeurIPS 2024 Workshop on New Frontiers in Adversarial Machine Learning
- HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal [\[arxiv\]](#)
[\[website\]](#)
Mantas Mazeika, Long Phan, **Xu Wang Yin**, Andy Zou, Zifan Wang, Norman Mu, Ellie Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, Dan Hendrycks
International Conference on Machine Learning (**ICML 2024**)
- End-to-End Signal Classification in Signed Cumulative Distribution Transform Space [\[arxiv\]](#)
AMH Rubaiyat, Shiyong Li, **Xu Wang Yin**, Mohammad Shifat E Rabbi, Yan Zhuang, Gustavo K. Rohde
IEEE Transactions on Pattern Analysis and Machine Intelligence (**PAMI 2023**)
- A Digital Camera-based Eye Movement Assessment Method for NeuroEye Examination [\[link\]](#)
Mohamed Abul Hassan, **Xu Wang Yin***, Yan Zhuang*, Chad M Aldridge, Timothy McMurry, Andrew M Southerland, Gustavo K Rohde
IEEE Journal of Biomedical and Health Informatics, 2023
- Representation Engineering: A Top-Down Approach to AI Transparency [\[arxiv\]](#)
Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, **Xu Wang Yin**, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, Dan Hendrycks
arXiv preprint, 2023
- Testing Robustness Against Unforeseen Adversaries [\[arxiv\]](#) [\[github\]](#)
Max Kaufmann, Daniel Kang, Yi Sun, Steven Basart, **Xu Wang Yin**, Mantas Mazeika, Akul Arora, Adam Dziedzic, Franziska Boenisch, Tom Brown, Jacob Steinhardt, Dan Hendrycks
arXiv preprint, 2023
- Learning Energy-Based Models With Adversarial Training
Xu Wang Yin, Shiyong Li, Gustavo K. Rohde
European Conference on Computer Vision (**ECCV 2022**) [\[code\]](#) [\[arxiv\]](#) [\[poster\]](#) [\[video\]](#)
- Radon Cumulative Distribution Transform Subspace Modeling for Image Classification
M Shifat-E-Rabbi, **Xu Wang Yin**, Abu Hasnat Mohammad Rubaiyat, Shiyong Li, Soheil Kolouri, Akram Aldroubi, Jonathan M. Nichols, Gustavo K. Rohde
Journal of Mathematical Imaging and Vision, 2021 [\[code\]](#)[\[Link\]](#)
- GAT: Generative Adversarial Training for Adversarial Example Detection and Robust Classification
Xu Wang Yin, Soheil Kolouri, Gustavo K. Rohde
International Conference on Learning Representations (**ICLR 2020**) [\[code\]](#)[\[arxiv\]](#)
- Neural Networks, Hypersurfaces, and Radon Transforms
Soheil Kolouri, **Xu Wang Yin**, Gustavo K. Rohde
IEEE Signal Processing Magazine, 2020 [\[code\]](#)[\[IEEEExplore link\]](#)[\[arxiv\]](#)
- OBJ2TEXT: Generating Visually Descriptive Language from Object Layouts [\[code\]](#)[\[Link\]](#)
Xu Wang Yin and Vicente Ordonez
Empirical Methods in Natural Language Processing (**EMNLP 2017**, oral presentation)

- Robust Text Detection in Natural Scene Images
Xu-Cheng Yin, **Xuwang Yin**, Kaizhu Huang, Hong-Wei Hao
IEEE Transactions on Pattern Analysis and Machine Intelligence (**PAMI 2014**) [[link](#)]
- Effective text localization in natural scene images with MSER, geometry-based grouping and AdaBoost
Xuwang Yin, Xu-Cheng Yin, Hong-Wei Hao, Khalid Iqbal
International Conference on Pattern Recognition (**ICPR 2012**) [[IEEEExplore link](#)]

Awards

- [Top Reviewer of NeurIPS 2022](#)
- International Conference on Document Analysis and Recognition (ICDAR2013) robust reading competition [winner](#) (out of 15 research teams from 7 countries)
- Outstanding Graduate of Beijing, Beijing Municipal Commission of Education, Jan 2014.

Professional Service

ACL-IJCNLP 2021, EMNLP 2021, ACL 2022, ICLR 2022, ECCV 2022, NeurIPS 2022, NeurIPS 2024, ACL 2024, NeurIPS 2025, ICLR 2026, Neurocomputing journal